A2 · Unit 5 · WCH15 Transition Metals and Organic Nitrogen Chemistry

Organic synthesis

This topic pulls the organic chemistry of the course together. It shows how to deduce a structure from data, how to plan a route from one compound to another, how to lengthen a carbon chain with Grignard reagents, and the practical techniques used to make, separate, purify and identify an organic product safely.

SPEC 20.1 Deducing formulae and structures from data

WHAT THIS SPEC POINT ASKS

Deduce empirical, molecular and structural formulae from combustion analysis, percentage composition, functional-group tests, infrared spectra, mass spectra and ^{13}C and ^{1}H NMR spectra.

A four-stage method

1. **Empirical formula** from combustion data or percentage composition.
2. **Molecular formula** from M_r (usually the M^+ peak in the mass spectrum).
3. **Functional groups** from chemical tests (19.12) and the IR spectrum.
4. **Full structure** from the NMR spectra and mass-spectrum fragments. Check that your structure fits **every** piece of data.

Stage 1a: combustion analysis

A known mass of the compound (C, H and maybe O) is burned in excess oxygen. The CO_2 and H_2O are collected and weighed.

- mass of C = mass of CO_2 × **12 ÷ 44**
- mass of H = mass of H_2O × **2 ÷ 18** (each H_2O contains **two** H atoms)
- mass of O = mass of sample − mass of C − mass of H (by difference)
- Then convert each mass to moles and find the simplest whole-number ratio.

Empirical formula from combustion

1. 0.290 g of a compound of C, H and O burns completely to give 0.660 g of CO_2 and 0.270 g of H_2O. Find its empirical formula.

2. Carbon: moles of CO_2 = 0.660 ÷ 44.0 = 0.0150 mol, so 0.0150 mol of C, mass 0.180 g.

3. Hydrogen: moles of H_2O = 0.270 ÷ 18.0 = 0.0150 mol, so 2 × 0.0150 = 0.0300 mol of H, mass 0.030 g.
4. Oxygen by difference: 0.290 − 0.180 − 0.030 = 0.080 g, so 0.080 ÷ 16.0 = 0.0050 mol.
5. Ratio C : H : O = 0.0150 : 0.0300 : 0.0050 = 3 : 6 : 1, so the empirical formula is **C_3H_6O**. If the mass spectrum shows M^+ at 58, this is also the molecular formula: propanone or propanal, which IR, NMR or chemical tests can tell apart.

WORKED EXAMPLE

Worked example: combustion analysis

1.44 g of compound Q (C, H and O only) burns completely to give 3.52 g of CO_2 and 1.44 g of H_2O.

Mass of C = 3.52 × 12 ÷ 44 = 0.960 g. Mass of H = 1.44 × 2 ÷ 18 = 0.160 g.

Mass of O = 1.44 − 0.960 − 0.160 = 0.320 g.

Moles: C = 0.960 ÷ 12 = 0.0800; H = 0.160 ÷ 1 = 0.160; O = 0.320 ÷ 16 = 0.0200.

Divide by the smallest (0.0200): C 4, H 8, O 1. Empirical formula **C_4H_8O** (M_r 72).

The molecular ion is at m/z 72, so the molecular formula is also **C_4H_8O**.

Stage 1b: percentage composition

WORKED EXAMPLE

Worked example: from percentages

A compound contains 40.0% C, 6.7% H and 53.3% O by mass, and its M_r is 60.

Moles in 100 g: C 40.0 ÷ 12 = 3.33; H 6.7 ÷ 1 = 6.7; O 53.3 ÷ 16 = 3.33.

Ratio 1 : 2 : 1, so the empirical formula is **CH_2O** (M_r 30).

60 ÷ 30 = 2, so the molecular formula is **$C_2H_4O_2$** (e.g. ethanoic acid or methyl methanoate).

Stage 3: infrared spectrum

Look up wavenumbers in the **Data Booklet** and quote a value **within its range**. The ones used most:

Bond	Wavenumber / cm^{-1}	Notes
O–H (alcohol, phenol)	3750–3200	broad if hydrogen-bonded
O–H (carboxylic acid)	3300–2500	very broad
N–H (amine)	3500–3300	
C–H (aldehyde)	2900–2820 and 2775–2700	two weak peaks
C≡N (nitrile)	2260–2215	
C=O (ester)	1750–1735	
C=O (aldehyde)	1740–1720	
C=O (ketone)	1720–1700	aryl ketone 1700–1680
C=O (carboxylic acid)	1725–1700	
C=C (alkene)	1669–1645	
C=C (arene ring)	about 1600, 1580, 1500, 1450	

Stage 4: mass spectrum

- The **molecular ion peak M^+** (the last major peak) gives the M_r.
- **Fragments** show pieces of the structure. Write them **with a + charge**: m/z 15 CH_3^+, 29 $C_2H_5^+$ (or CHO^+), 43 CH_3CO^+ (or $C_3H_7^+$), 57 $C_2H_5CO^+$, 77 $C_6H_5^+$.
- **Chlorine** has isotopes ^{35}Cl : ^{37}Cl = 3 : 1. One Cl gives peaks at M and M+2 in a **3 : 1** ratio; two Cl give M, M+2 and M+4 in a **9 : 6 : 1** ratio.
- A small **M+1** peak comes from molecules containing one ^{13}C atom.

Stage 4: NMR spectra

^{13}C NMR: the number of peaks = the number of **different carbon environments**. Look for symmetry: in 1,4-disubstituted benzene the ring has fewer environments than in the 1,2- or 1,3-isomers.

^1H NMR gives three kinds of information:

Feature	Tells you
number of peaks	number of different H environments
chemical shift δ / ppm	the type of environment (use the Data Booklet). Roughly: alkyl 0.8–1.5; H–C next to C=O 2.0–2.8; H–C–O 3.3–4.3; arene H 6.5–8.0; aldehyde H 9–10; COOH 10–12

Feature	Tells you
relative peak area (**integration**)	the ratio of H atoms in each environment
splitting	**n + 1 rule**: a peak is split into (n + 1) lines by n H atoms on the **neighbouring carbon(s)**. Singlet = 0 neighbours, doublet = 1, triplet = 2, quartet = 3

- An **O–H** (or N–H) proton usually gives a **singlet**: it neither splits nor is split.
- Equivalent H atoms don't split each other.

WORKED EXAMPLE

Worked example: putting the evidence together

Q is C_4H_8O (from the combustion example above).

- Tests: an **orange precipitate** with 2,4-DNPH (C=O); **no** silver mirror with Tollens' reagent (not an aldehyde), so it is a **ketone**. A **yellow precipitate** with iodine and NaOH shows $CH_3CO–$.
- IR: strong peak at 1715 cm^{-1} (ketone C=O); no broad O–H.
- Mass spectrum: M^+ at 72; fragments at 43 (CH_3CO^+) and 57 ($CH_3CH_2CO^+$).
- ^{13}C NMR: 4 peaks, so 4 carbon environments.
- ^1H NMR: three peaks, areas 3 : 2 : 3. A **singlet** at 2.1 ppm (CH_3 next to C=O, no H neighbours); a **quartet** at 2.4 ppm (CH_2, 3 H neighbours); a **triplet** at 1.0 ppm (CH_3, 2 H neighbours).
- Structure: $CH_3COCH_2CH_3$, **butanone**.

MARK-SCHEME WORDING

- Combustion: "mass of C = 4.95 × (12 ÷ 44)", "mass of H = 0.90 × (2 ÷ 18)", "mass of O by difference", then moles and ratio. A correct answer with working scores all marks (WCH15 Oct 2025, Jun 2022).
- Rounding moles: "carbons per molecule = 95.6 ÷ 24 = 3.98, so 4" (WCH15 Jun 2024).
- ^1H NMR: "4 peaks means 4 (different) hydrogen environments"; "peak at 7.5 ppm is due to H on the benzene ring"; quartet "3 H on the neighbouring carbon"; singlet "no H on neighbouring carbon"; "area 5 for the ring shows C_6H_5" (WCH15 Jan 2022).
- "The OH proton neither splits nor causes splitting" (WCH15 Jan 2021). Unusual wording such as "two splits" is accepted if clear.
- Chemical shifts and IR: "allow a single value or a range **within** the Data Booklet range". A value outside the range scores 0.
- Fragments need the **+ charge** ("penalise omission of the positive charge"); a radical instead of an ion is wrong.
- Isotopes: "two atoms of chlorine, so a 9 : 6 : 1 ratio"; M+1 "due to one ^{13}C atom" (WCH15 Jun 2024).

Often asked as: empirical formula from combustion data (most series); a 6–7 mark "deduce the structure" from combined data; count ^{13}C environments; assign ^{1}H NMR peaks with splitting; identify fragment ions.

COMMON MISTAKE

Taking the moles of H_2O as the moles of H. That **halves** the hydrogen (Jun 2022 report: the commonest error).

Using trial and error instead of the systematic C, H, O route (Jan 2022 report).

Suggesting N-containing groups for a compound with no N in its formula.

Counting the C=O carbon as a proton environment, or counting both halves of a symmetrical molecule as different (Jan 2021, Oct 2024 reports).

Quoting IR or ^{13}C data when the question asked about ^{1}H NMR (Oct 2021 report).

Writing "^{36}Cl" or "$^{35.5}Cl$" (Jun 2024 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Inverse calculation**: percentage composition from a given formula, and values in ppm or ppb (WCH15 Oct 2022).
- **Gas chromatography and GC–MS**: components separate by retention time (how strongly each is attracted to the stationary phase), then each goes into the mass spectrometer (WCH15 Jun 2025).
- **Empirical formula from a skeletal formula** (about a third of candidates managed it in WCH15 Oct 2024). Count every carbon and hydrogen carefully.

SPEC 20.2 Lengthening the carbon chain: Grignard reagents

WHAT THIS SPEC POINT ASKS

Understand how to lengthen a carbon chain by using magnesium to make Grignard reagents, and how these react with carbon dioxide and with carbonyl compounds in dry ether.

Making a Grignard reagent

$$CH_3CH_2Br + Mg \rightarrow CH_3CH_2MgBr \text{ (ethylmagnesium bromide)}$$

- **Reagents and conditions**: a halogenoalkane (or halogenoarene, e.g. bromobenzene) with **magnesium** turnings, in **dry ether** (ethoxyethane), warmed gently under reflux.

- **Why dry?** Grignard reagents react with water at once, giving an alkane: $CH_3CH_2MgBr + H_2O \rightarrow CH_3CH_3 +$ Mg(OH)Br. All apparatus and reagents must be **anhydrous**.
- The C–Mg bond is polar with the carbon **δ−** (Mg is much less electronegative). So the carbon acts as a **nucleophile**: it attacks δ+ carbon atoms and forms a **new C–C bond**.

Reactions (two stages each: add, then dilute acid)

1. The Grignard reagent adds to the δ+ carbon of C=O (CO_2, an aldehyde or a ketone) in dry ether.
2. **Dilute acid** (e.g. dilute HCl or H_2SO_4) is added to hydrolyse the magnesium salt and release the product.

Grignard reacts with	Product (after dilute acid)	Example with CH_3CH_2MgBr
carbon dioxide (dry ice)	**carboxylic acid**, one C longer	propanoic acid, CH_3CH_2COOH
methanal	**primary** alcohol	propan-1-ol, $CH_3CH_2CH_2OH$
any **other aldehyde**	**secondary** alcohol	with ethanal: butan-2-ol, $CH_3CH_2CH(OH)CH_3$
a **ketone**	**tertiary** alcohol	with propanone: 2-methylbutan-2-ol, $CH_3CH_2C(OH)(CH_3)_2$

Equations, e.g.:

- $CH_3CH_2MgBr + CO_2 \rightarrow CH_3CH_2COOMgBr$, then $CH_3CH_2COOMgBr + H^+ \rightarrow CH_3CH_2COOH + Mg^{2+} + Br^-$
- $CH_3CH_2MgBr + CH_3CHO \rightarrow CH_3CH_2CH(CH_3)OMgBr$, then $+ H_2O \rightarrow CH_3CH_2CH(OH)CH_3 +$ Mg(OH)Br

Planning a Grignard route (working backwards)

1. Find the **carbon carrying the OH** (or the COOH carbon).
2. That carbon was the **C=O carbon** (or the CO_2 carbon) of the starting carbonyl.
3. **One** of the groups attached to it came from the Grignard reagent. Split the molecule there.

WORKED EXAMPLE

Worked example: making 2-methylbutan-2-ol, $(CH_3)_2C(OH)CH_2CH_3$

The OH carbon has two CH_3 groups and one CH_2CH_3 group, so it is a tertiary alcohol: make it from a **ketone**.

Route 1: CH_3CH_2MgBr + **propanone** (CH_3COCH_3), then dilute acid.

Route 2: CH_3MgBr + **butanone** ($CH_3COCH_2CH_3$), then dilute acid.

Each Grignard is made from the bromoalkane + Mg in **dry ether**.

Other ways to add carbon atoms

Method	Reagents	Gives	Carbons added
Grignard + CO_2	RMgX, then CO_2, then H^+	carboxylic acid	1
Grignard + carbonyl	RMgX + aldehyde or ketone, then H^+	alcohol	the R group
Cyanide + halogenoalkane	KCN in ethanol, reflux	nitrile (then acid or amine)	1
Cyanide + carbonyl	HCN + KCN	hydroxynitrile	1
Friedel–Crafts on an arene	RCl or RCOCl + $AlCl_3$	alkyl- or acylbenzene	R or RCO

MARK-SCHEME WORDING

- "magnesium (powder) and (**dry**) ether"; "structure of the Grignard reagent" CH_3CH_2MgBr (CH_3CH_2BrMg is not accepted); "reaction with CO_2" then "hydrolysis with dilute acid / H^+". CO_2 and acid can be two labelled parts of one arrow.
- Missing "dry" ether is penalised (WCH15 Jan 2024).
- Grignard + methanal → primary alcohol; + other aldehyde → secondary; + ketone → tertiary; + CO_2 → carboxylic acid (WCH15 Oct 2020 MCQ). The Grignard acts as a **nucleophile** (WCH15 Jun 2024 MCQ).
- Dehydrating a Grignard alcohol product: "concentrated phosphoric acid or concentrated sulfuric acid" (or alumina) (WCH15 Jan 2021).

Often asked as: a 5–7 mark synthesis route using a Grignard step (benzene → benzoic acid, WCH15 Jan 2022; alcohol → acid one carbon longer, WCH15 Jan 2025); which Grignard and carbonyl give a named alcohol (MCQ).

COMMON MISTAKE

Forgetting **dry** ether, or merging CO_2 and the acid hydrolysis into one step with no acid (Jan 2025 report).

Making the Grignard from a compound that still has an –OH group: the OH would destroy it (Oct 2024 report).

Wrong number of carbons. Count before and after every step.

Putting Br on the ring as well as Mg in the aryl Grignard. It is C_6H_5MgBr: the Mg goes **between** the ring carbon and the Br (Jan 2022 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Grignard reagents and $LiAlH_4$ both react with water, which is why both need dry conditions (WCH15 Jan 2025 report). Choosing between a Grignard route and a cyanide route to the same acid is also examined (WCH15 Jun 2022).

SPEC 20.3 Problem solving in organic chemistry

WHAT THIS SPEC POINT ASKS

Use the organic chemistry in the specification to: (i) predict the properties of unfamiliar compounds with one or more known functional groups; (ii) plan reaction schemes of up to four steps; (iii) choose suitable practical procedures; (iv) choose control measures to reduce risk, using hazard data.

(i) Predicting properties of an unfamiliar compound

Find **each functional group** and apply its chemistry one at a time. Then check for things the groups do together (isomerism, polymers, hydrogen bonding).

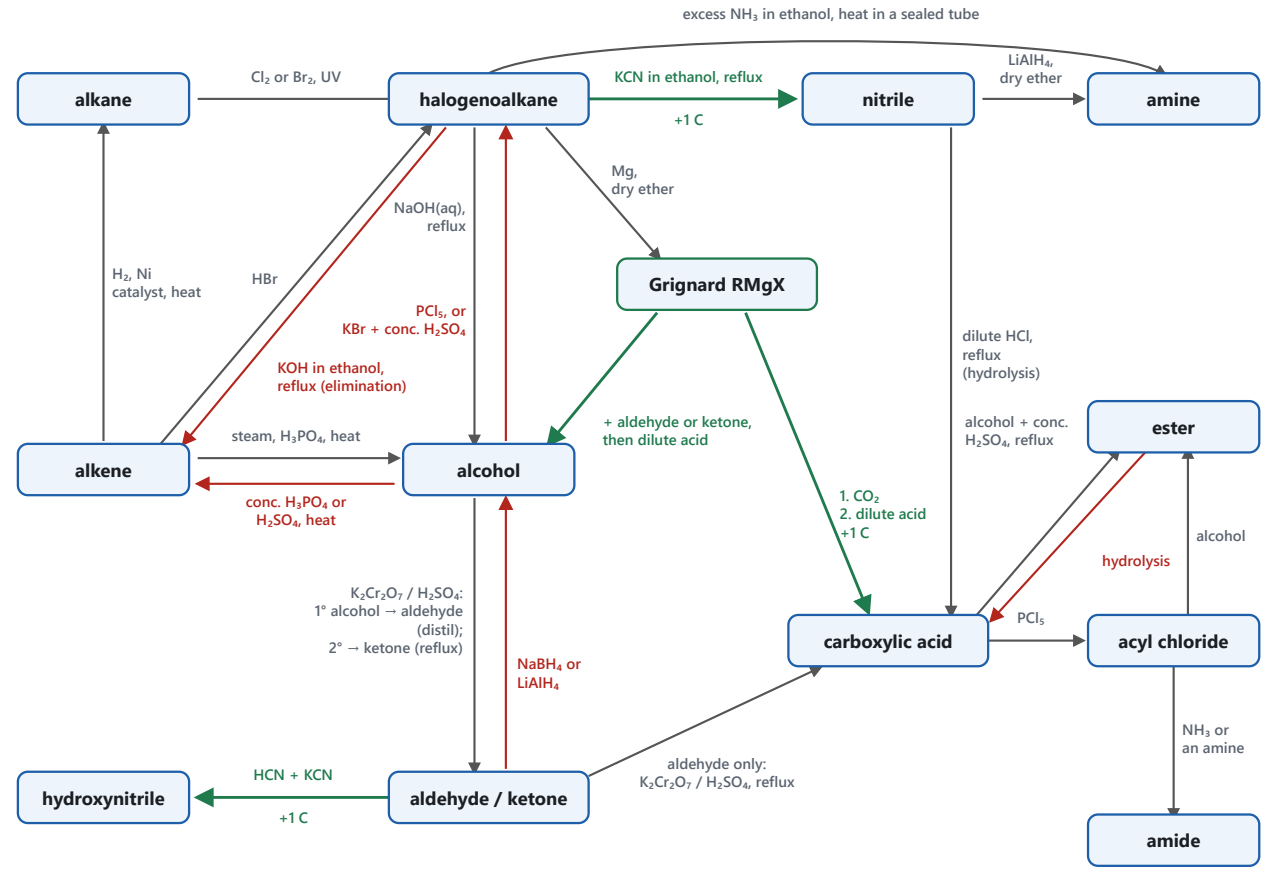
WORKED EXAMPLE

Worked example: predicting the reactions of $CH_2{=}CHCH_2CH(OH)COOH$

The compound is 2-hydroxypent-4-enoic acid. Its groups: an **alkene**, a **secondary alcohol** and a **carboxylic acid**.

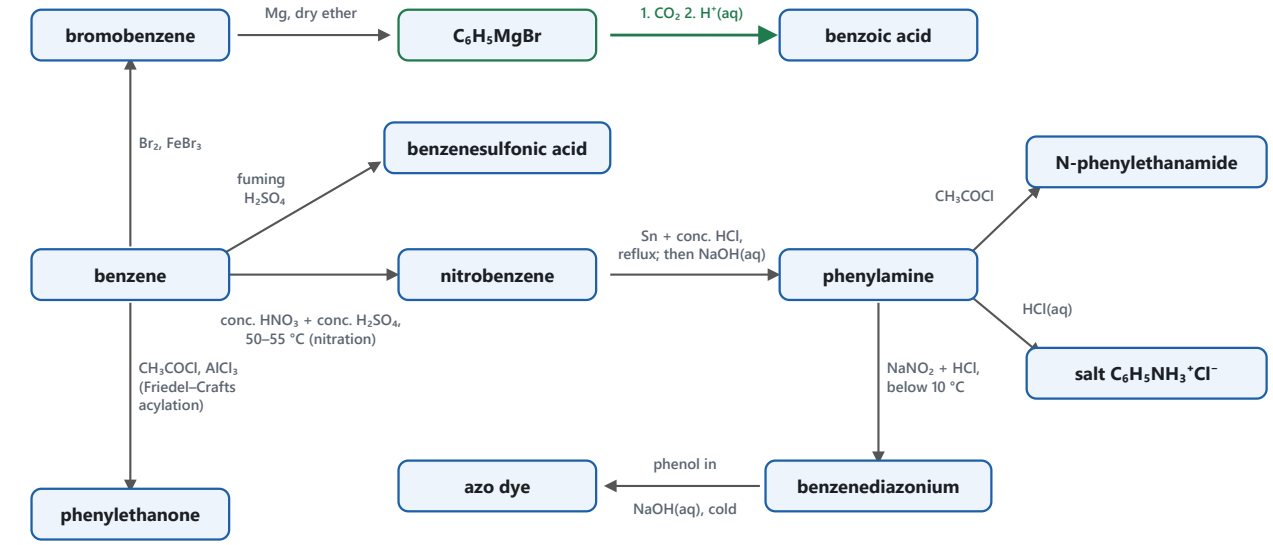
- **Bromine water**: decolourises (C=C addition), giving $CH_2BrCHBrCH_2CH(OH)COOH$.
- **Na_2CO_3(aq)**: fizzing, CO_2 given off (COOH).
- **PCl_5**: steamy fumes of HCl (both OH and COOH react).
- **Acidified $K_2Cr_2O_7$, warm**: orange → green. The secondary alcohol is oxidised to a ketone.
- **Iodine + NaOH**: no yellow precipitate. There is no $CH_3CH(OH)$– or CH_3CO– group.
- **Isomerism**: C2 is chiral (H, OH, COOH and $CH_2CH{=}CH_2$), so it has optical isomers. No E/Z isomers: the end carbon of the C=C has two H atoms.
- **Polymers**: OH and COOH in one molecule, so it can form a **polyester** by condensation; the C=C allows **addition** polymerisation too.
- **Solubility**: soluble in water, because the OH and COOH groups hydrogen-bond with water.

(ii) Planning a synthesis (up to four steps)



Green arrows add a carbon atom. Red arrows are the reverse of the black arrow beside them.

Aliphatic reaction map. Green arrows add a carbon atom.



Also: benzene + RCl and $AlCl_3$ → alkylbenzene (alkylation). Phenol + Br_2(aq) → 2,4,6-tribromophenol (no catalyst).

Aromatic reaction map (Topics 18 and 19).

Method

1. Write out the **start** and the **target**. Compare the **number of carbons** and the **functional groups**.
2. If the chain must get longer, you need a **cyanide** or **Grignard** step (or Friedel–Crafts on a ring). If it must get shorter by one, use the **iodoform** reaction (beyond the spec).
3. **Work backwards** from the target: "what could I make this from?" Then work forwards from the start, and find where the two meet.
4. For **every step** give the **reagent(s)**, the **conditions** (heat/reflux, distil, catalyst, solvent, temperature) and the **intermediate's structure**.

Reagents and conditions: master table

Change	Reagents and conditions
alkane → halogenoalkane	Cl_2 or Br_2, UV light
alkene → alkane	H_2, Ni catalyst, heat
alkene → halogenoalkane	HBr (or HCl), room temperature
alkene → dibromoalkane	Br_2, room temperature
alkene → alcohol	steam, H_3PO_4 catalyst, high temperature and pressure
halogenoalkane → alcohol	NaOH(aq) or KOH(aq), heat under reflux
halogenoalkane → alkene	KOH in ethanol, heat under reflux
halogenoalkane → nitrile	KCN in ethanol, heat under reflux (+1 C)
halogenoalkane → primary amine	excess NH_3 in ethanol, heat in a sealed tube
halogenoalkane → Grignard	Mg, dry ether
alcohol → chloroalkane	PCl_5, room temperature, dry
alcohol → bromoalkane	KBr + concentrated H_2SO_4 (makes HBr), heat
alcohol → alkene	concentrated H_3PO_4 or H_2SO_4, heat (or Al_2O_3)
1° alcohol → aldehyde	$K_2Cr_2O_7$ + dilute H_2SO_4, warm, **distil off** the aldehyde as it forms
1° alcohol → carboxylic acid	excess $K_2Cr_2O_7$ + dilute H_2SO_4, heat under **reflux**
2° alcohol → ketone	$K_2Cr_2O_7$ + dilute H_2SO_4, heat under reflux
aldehyde or ketone → alcohol	$NaBH_4$ (in water or methanol), or $LiAlH_4$ in dry ether

Change	Reagents and conditions
aldehyde or ketone → hydroxynitrile	HCN + KCN (+1 C)
carboxylic acid → primary alcohol	$LiAlH_4$ in dry ether, then dilute acid
carboxylic acid → acyl chloride	PCl_5 (or $SOCl_2$), dry
acid + alcohol → ester	concentrated H_2SO_4 catalyst, heat under reflux
acyl chloride → ester / amide / acid	alcohol / NH_3 or amine / water, room temperature
ester → acid + alcohol	dilute acid, reflux (or NaOH(aq), reflux, gives the carboxylate)
nitrile → carboxylic acid	dilute HCl, heat under reflux
nitrile → primary amine	$LiAlH_4$ in dry ether
benzene → nitrobenzene	conc. HNO_3 + conc. H_2SO_4, 50–55 °C
nitrobenzene → phenylamine	Sn + conc. HCl, reflux; then NaOH(aq)
phenylamine → diazonium ion	$NaNO_2$ + HCl, below 10 °C

WORKED EXAMPLE

Worked example: ethanol to propanoic acid (one more carbon)

Ethanol has 2 C and propanoic acid has 3 C, so a chain-lengthening step is needed.

Route A (cyanide):

1. Ethanol → bromoethane: KBr + concentrated H_2SO_4, heat.
2. Bromoethane → propanenitrile, CH_3CH_2CN: KCN in ethanol, heat under reflux.
3. Propanenitrile → propanoic acid: dilute HCl, heat under reflux.

Route B (Grignard): steps 1 as above; then Mg in dry ether → CH_3CH_2MgBr; then CO_2, followed by dilute acid → CH_3CH_2COOH.

WORKED EXAMPLE

Worked example: benzene to N-phenylethanamide, with overall yield

1. Benzene → nitrobenzene: conc. HNO_3 + conc. H_2SO_4, 50–55 °C. Yield 70%.
2. Nitrobenzene → phenylamine: Sn + conc. HCl, reflux, then NaOH(aq). Yield 60%.

3. Phenylamine → N-phenylethanamide ($C_6H_5NHCOCH_3$, M_r 135): ethanoyl chloride, room temperature. Yield 85%.

Overall yield = 0.70 × 0.60 × 0.85 = 0.357 = **35.7%**. Multiply the decimal yields; never add them.

What volume of benzene (M_r 78, density 0.879 g cm^{-3}) is needed for 5.00 g of product?

Moles of product = 5.00 ÷ 135 = 0.03704 mol.

Moles of benzene needed = 0.03704 ÷ 0.357 = 0.1037 mol. **Divide** by the yield: you need more than the theoretical amount.

Mass = 0.1037 × 78 = 8.09 g. Volume = 8.09 ÷ 0.879 = **9.21 cm^3**.

(iii) Choosing a procedure

Situation	Procedure and why
slow reaction with volatile liquids	**heat under reflux**: heats for a long time without losing material
product must be removed as it forms (aldehyde from a 1° alcohol)	**distil** it off, so it isn't oxidised further
flammable solvent or reactant (ether, ethanol)	**water bath or electric heating mantle**, no naked flame
unstable intermediate (diazonium ion), exothermic nitration	**ice bath**; add reagents slowly
Grignard, $LiAlH_4$, $AlCl_3$, PCl_5, acyl chlorides	**dry** apparatus and reagents (react with water)
only one group should react	a **selective reagent**: H_2/Ni reduces C=C but not C=O; $NaBH_4$ reduces C=O but not C=C

(iv) Hazards and control measures

Hazard in the data	Control measure
flammable	no naked flames; heat with a water bath or electric heating mantle
toxic or harmful vapour, or a toxic gas formed (HCl, HCN, Br_2)	**fume cupboard**
corrosive (conc. acids, PCl_5, acyl chlorides, NaOH)	gloves and eye protection; add slowly

Hazard in the data	Control measure
irritant	gloves, eye protection; avoid skin contact
carcinogen (benzene)	use a less harmful substitute (methylbenzene or methoxybenzene) if possible, or work in a fume cupboard with gloves
reacts violently with water	keep dry; add carefully, in small portions

Control measures must **match the hazard**: "wear goggles" doesn't help with a flammable solvent.

KEY IDEA

Model 6-mark points: ethanoyl chloride v ethanoic anhydride for acylating phenylamine (WCH15 Jan 2024).

- Ethanoyl chloride gives **HCl**, a toxic, corrosive gas; the anhydride gives **ethanoic acid**, which is less hazardous (though flammable).
- Ethanoyl chloride reacts **violently** (with water too), so it is **harder to control**; the anhydride reacts more **slowly**.
- Use the hazard information the question gives for each reagent.
- **Atom economy** for making N-phenylethanamide (M_r 135):
  - with ethanoyl chloride: 135 ÷ (93 + 78.5) × 100 = **78.7%**
  - with ethanoic anhydride: 135 ÷ (93 + 102) × 100 = **69.2%**
- A higher atom economy means less waste: an advantage of the acyl chloride.

MARK-SCHEME WORDING

- Oxidation: "$K_2Cr_2O_7$ and H_2SO_4 / acidified dichromate"; aldehyde "with distillation" (reflux is not accepted for the aldehyde); acid "reflux". Missing H^+, or HCl used for the acid, is penalised once (WCH15 Jan 2024, Oct 2024).
- Reduce only C=C: "H_2 and Ni / Pt". $LiAlH_4$ is not accepted, because it would reduce the ketone too (WCH15 Jan 2024, Oct 2024).
- Intermediates score even if a reagent is wrong. Give only **one** route: if you give two, the one scoring more marks is credited (WCH15 Jan 2024).
- Comparing two routes, any three: "number of steps", "purification required", "availability / cost of reagents", "toxicity of reagents / intermediates / by-products", "atom economy", "yield of each step", "side reactions". Unjustified "cost", "safety", "yield" or "rate" are ignored (WCH15 Jun 2024).

Often asked as: a 4–7 mark synthesis (almost every series); choose a selective reagent; a 6-mark hazards-and-atom-economy comparison; reverse yield calculations; pick a hazard symbol.

COMMON MISTAKE

Not knowing where to start. Work **backwards** from the target (Oct 2020, Jun 2022 reports).

Leaving out the acid catalyst in esterification (Jun 2022 report: the commonest slip).

Adding cyanide to benzene as a first step: that reaction doesn't happen (Oct 2023 report).

Oxidising a primary alcohol under reflux when you need the aldehyde. Distil it off (Oct 2024 report).

Hazard answers that just say "safety" or "harmful", with no atom-economy numbers (Jan 2024 report). Show the molar masses.

Multiplying by the percentage yield in a reverse calculation. **Divide** (Jun 2022 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Descending the series**: the iodoform reaction (I_2 + NaOH) turns $CH_3CO–R$ into $RCOO^-$ + CHI_3, removing one carbon (WCH15 Jan 2021).
- **Unfamiliar mechanisms** given in the stem (radical steps with half-arrows, additions to C=O, catalytic cycles): follow the pattern shown and keep the arrows from bonds or lone pairs (WCH15 Oct 2021, Jan 2026).
- **Industrial context**: a factory route can't use some lab reagents (cost, hazard, scale), and more steps usually means a lower overall yield (WCH15 Jun 2025).

SPEC 20.4 Core Practical 16: preparation of aspirin

WHAT THIS SPEC POINT ASKS

Core Practical 16: prepare aspirin, and purify it.

The reaction

2-Hydroxybenzoic acid (salicylic acid) + ethanoic anhydride → 2-ethanoyloxybenzoic acid (aspirin) + ethanoic acid

$$HOC_6H_4COOH + (CH_3CO)_2O \rightarrow CH_3COOC_6H_4COOH + CH_3COOH$$

- The **phenol –OH** group of salicylic acid is **acylated**, forming an **ester**. Phenols react only slowly with carboxylic acids, so a more reactive **acid anhydride** is used. An **acid catalyst** (a few drops of concentrated H_2SO_4 or 85% H_3PO_4) speeds it up.
- Aspirin contains an **ester** group, a **carboxylic acid** group and a **benzene ring**.

CORE PRACTICAL

Core practical 16: method in brief

Preparation

1. Weigh about 2–3 g of 2-hydroxybenzoic acid into a dry conical (or pear-shaped) flask. Record the exact mass.
2. In a **fume cupboard**, add about 4–5 cm³ of **ethanoic anhydride** (in **excess**) and 5 drops of concentrated sulfuric or phosphoric acid.
3. Warm in a **water bath** at about 60–80 °C for about 15–20 minutes, swirling, until the solid dissolves.
4. Carefully add a little **cold water** to react with (hydrolyse) the excess ethanoic anhydride, then more cold water. Cool in an **ice bath** until crystals form.
5. Collect the crude crystals by **filtration under reduced pressure** (Büchner funnel) and wash with a little **ice-cold water**.

Purification by recrystallisation

6. Dissolve the crude product in the **minimum volume of hot solvent** (hot water, or a little hot ethanol then warm water).
7. If there are insoluble bits, filter while **hot**.
8. Let the solution **cool slowly**, then in ice, so that pure crystals form. Soluble impurities stay in solution.
9. Filter under reduced pressure, wash with a little ice-cold water, and **dry** (between filter papers or in a desiccator). Weigh.

Purity check: measure the **melting temperature**. Pure aspirin melts sharply at about 135 °C; an impure sample melts lower and over a wider range.

Treating the results

WORKED EXAMPLE

Worked example: percentage yield and limiting reagent

3.00 g of 2-hydroxybenzoic acid (M_r 138) reacts with 4.00 cm³ of ethanoic anhydride (M_r 102, density 1.08 g cm^{-3}). 2.85 g of pure aspirin (M_r 180) is obtained.

Moles of acid = 3.00 ÷ 138 = 0.02174 mol.

Mass of anhydride = 4.00 × 1.08 = 4.32 g; moles = 4.32 ÷ 102 = 0.04235 mol. Ratio 1 : 1, so the anhydride is in excess and **2-hydroxybenzoic acid is limiting**.

Theoretical mass of aspirin = 0.02174 × 180 = 3.913 g.

Percentage yield = 2.85 ÷ 3.913 × 100 = **72.8%**.

WORKED EXAMPLE

Worked example: atom economy

Atom economy = M_r of the desired product ÷ total M_r of all the reactants (from the balanced equation) × 100.

With ethanoic anhydride: 180 ÷ (138 + 102) × 100 = **75.0%** (by-product ethanoic acid).

With ethanoyl chloride (by-product HCl): 180 ÷ (138 + 78.5) × 100 = **83.1%**.

The anhydride is still chosen: it is cheaper, less violent, and doesn't give off corrosive HCl gas.

Why the yield is less than 100%

- Product is lost on transfer, and on the filter paper and glassware.
- Some aspirin stays **dissolved in the solvent** during recrystallisation and washing.
- The reaction may be incomplete, or some aspirin may be hydrolysed back to salicylic acid.

Sources of error and improvements

- Product still **damp** when weighed gives a mass that is too high. Dry it to constant mass.
- Too much recrystallisation solvent lowers the yield. Use the **minimum** volume of hot solvent.
- Washing with warm water dissolves product. Use **ice-cold** water, in small amounts.

MARK-SCHEME WORDING

- Equation: "salicylic acid + ethanoic anhydride → acetylsalicylic acid + ethanoic acid" (skeletal accepted; molecular formulae not) (WCH15 Oct 2023).
- % yield: 2.00 g salicylic acid (M_r 138) gives a theoretical 2.6087 g of aspirin (M_r 180); 1.9513 g obtained, so 74.8% (WCH15 Oct 2023).
- Recrystallisation: "dissolve in the **minimum** volume of **hot** solvent, filter (hot) to remove **insoluble** impurities, cool, and filter to collect crystals". Soluble impurities stay in the solution. The solvent must dissolve the product well **when hot** and poorly **when cold** (WCH15 Jan 2025, Oct 2022 MCQs).
- Melting-point bath: **mineral oil**, because water boils below aspirin's melting temperature. The likely impurity is unreacted **2-hydroxybenzoic acid** (WCH15 Jun 2024 MCQs).

Often asked as: the equation, percentage yield, recrystallisation and melting-point MCQs, and a whole Section C on aspirin (WCH15 Oct 2023, 19 marks).

COMMON MISTAKE

Using the anhydride (in excess) for the theoretical yield. Use the **limiting** reagent.

Using M_r values for the wrong compound. The M_r of aspirin is 180; salicylic acid is 138.

Forgetting ethanoic acid when balancing the equation (Oct 2023 report).

Naming 2-hydroxybenzoic acid wrongly: only a quarter of candidates got it (Oct 2023 report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Back titration for aspirin content**: each aspirin uses **2 NaOH** (one to neutralise the COOH, one to hydrolyse the ester) (WCH15 Oct 2023).
  - A 0.330 g tablet is heated with 25.0 cm^3 of 0.200 mol dm^{-3} NaOH (5.00×10^{-3} mol). The excess NaOH needs 16.70 cm^3 of 0.100 mol dm^{-3} HCl (1.670×10^{-3} mol).
  - NaOH used by aspirin = $5.00 \times 10^{-3} - 1.670 \times 10^{-3} = 3.330 \times 10^{-3}$ mol. Aspirin = $3.330 \times 10^{-3} \div 2 = 1.665 \times 10^{-3}$ mol.
  - Mass = $1.665 \times 10^{-3} \times 180 = 0.2997$ g. Percentage of the tablet = $0.2997 \div 0.330 \times 100$ = **90.8%**.
- **Aspirin in the body**: it is a weak acid, so in the acidic stomach (extra H^+) its dissociation equilibrium shifts **left**, and in the alkaline small intestine (OH^- removes H^+) it shifts **right**. Don't say K_a changes (WCH15 Oct 2023).
- **Oil of wintergreen** (methyl 2-hydroxybenzoate) is made from 2-hydroxybenzoic acid and **methanol** with an acid catalyst. In ^1H NMR, aspirin's COOH proton is at 10–12 ppm (WCH15 Oct 2023).
- **Iron(III) chloride** test: a purple colour shows unreacted phenol (salicylic acid) left in the product.

SPEC 20.5 Techniques for preparing and purifying organic compounds

WHAT THIS SPEC POINT ASKS

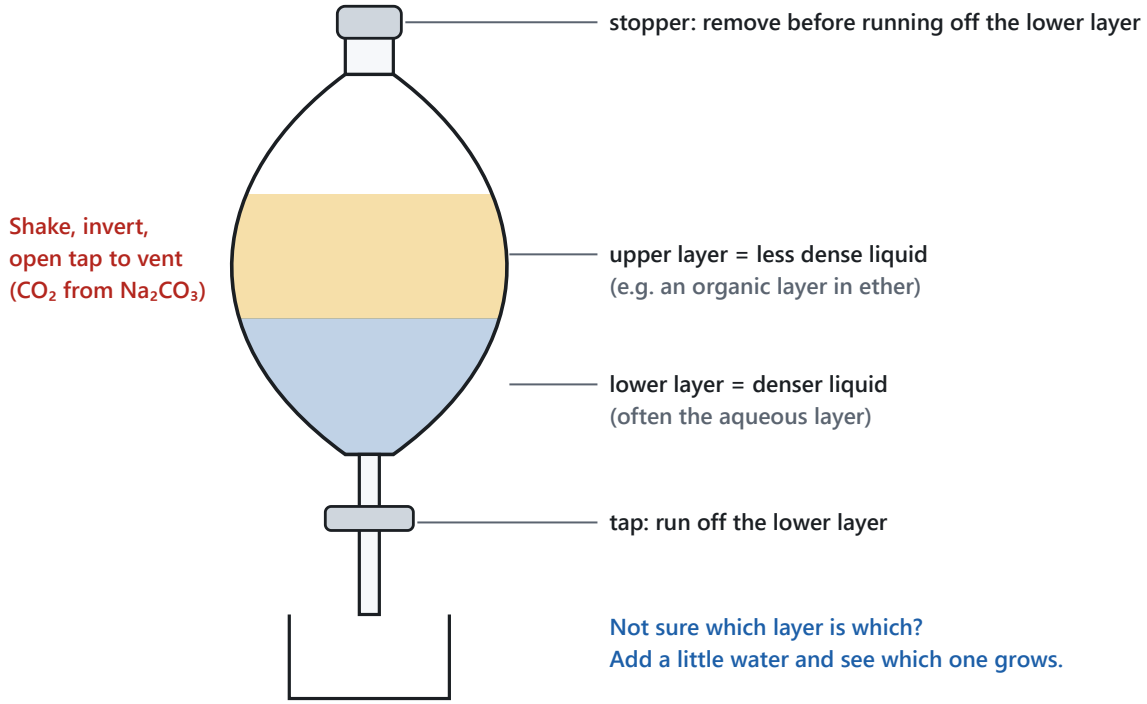
Understand these techniques: (i) refluxing; (ii) washing, including with water and sodium carbonate solution; (iii) solvent extraction; (iv) recrystallisation; (v) drying; (vi) distillation; (vii) steam distillation; (viii) melting temperature determination; (ix) boiling temperature determination.

(i) Heating under reflux

- A round-bottomed or pear-shaped flask with **anti-bumping granules**, and a **vertical condenser** with its top **open**. Water enters the condenser at the **bottom** and leaves at the top.
- **Why**: vapours condense and drip back, so the mixture can be **heated for a long time** without losing volatile reactants or products. This lets slow organic reactions go to completion.
- Heat with an **electric heating mantle** or **water bath** (flammable liquids). Never stopper the top: pressure would build up.
- Anti-bumping granules make the liquid **boil smoothly**, without large bubbles.

(ii) Washing

- The crude organic liquid is shaken with an aqueous solution in a **separating funnel**, and the layers are separated.
- **Water** removes water-soluble impurities (e.g. acids, salts, small alcohols).
- **Sodium carbonate solution** (or $NaHCO_3$) **neutralises acidic impurities** (e.g. HCl, H_2SO_4, carboxylic acid), turning them into water-soluble salts. CO_2 is given off, so **invert the funnel and open the tap often** to release the pressure.
- Sometimes a NaCl solution (brine) wash follows, to remove most of the water.



Separating funnel. Run off the lower layer through the tap, then pour the upper layer out of the top.

(iii) Solvent extraction

- Shake the aqueous mixture with an **organic solvent that is immiscible with water** (e.g. ethoxyethane or dichloromethane) and in which the **product is much more soluble** than in water. The product moves into the organic layer.
- Separate the layers. Use **several small portions** of solvent rather than one large portion: this extracts more product.
- Combine the organic layers, dry them (v), and remove the solvent by distillation.
- Adding salt to the water ("**salting out**") lowers the organic compound's solubility in water and helps it move into the organic layer (used for phenylamine, 19.5).

(iv) Recrystallisation (purifying a solid)

1. Choose a solvent in which the product is **very soluble when hot** but **only slightly soluble when cold**.
2. Dissolve the impure solid in the **minimum volume of hot solvent**, so the solution is saturated when hot.
3. **Filter hot** (warmed funnel) to remove **insoluble** impurities.
4. **Cool slowly**: the product crystallises. **Soluble** impurities are present in small amounts, so they stay in solution.
5. Filter under **reduced pressure** (Büchner funnel), wash with a little **ice-cold** solvent, and dry.

Repeating recrystallisation gives a purer product but a lower yield.

(v) Drying

- **Liquids**: add an anhydrous drying agent such as **anhydrous magnesium sulfate, calcium chloride or sodium sulfate**. Swirl and add more until some powder moves freely and the liquid is **clear**, not cloudy. Then **decant or filter** off the solid.
- **Solids**: press between filter papers, leave in a **desiccator**, or warm in a low oven, well below the melting temperature.

(vi) Distillation

- Separates liquids by **boiling temperature**, or a volatile liquid from non-volatile substances.
- Thermometer **bulb level with the side arm** (measures the temperature of the vapour that is distilling). Condenser sloping down, water in at the lower end.
- Collect the fraction that distils **within about 2 °C of the expected boiling temperature**.
- Also used to **remove a product as soon as it forms**, e.g. an aldehyde from a primary alcohol, before it is oxidised further.

(vii) Steam distillation

- Steam is passed through the mixture (or it is boiled with water). A **volatile compound that is immiscible with water** distils over with the steam **below 100 °C**, well below its own boiling temperature.
- **Why**: the compound can be separated **without heating it to its boiling temperature**, where it might **decompose**. It also leaves behind non-volatile tars and inorganic salts.
- The distillate is cloudy: two layers of water and the compound. Separate them with a separating funnel or by solvent extraction.
- Used for **phenylamine** (from the tin/HCl mixture, 19.5) and to extract **essential oils** (e.g. lavender oil, limonene) from plants.

(viii) Melting temperature determination

- Pack a little **dry, finely powdered** solid into a capillary tube sealed at one end. Heat it in a melting-point apparatus or an **oil bath** (Thiele tube), **slowly** (about 1–2 °C per minute) near the melting temperature.
- Record the temperature range from first melting to completely liquid.

- **Pure**: melts **sharply** (over 1–2 °C) at the Data Book value. **Impure**: melts at a **lower** temperature, over a **wider range**.

(ix) Boiling temperature determination

- Distil the liquid (or heat it under reflux with a thermometer in the vapour). A **pure** liquid boils at a **constant temperature** that matches the Data Book value.
- Impurities make it boil **over a range** of temperatures. A dissolved non-volatile impurity **raises** the boiling temperature.

Technique	Removes or does
reflux	heats for a long time without loss of volatile substances
washing with water	water-soluble impurities
washing with Na_2CO_3(aq)	acidic impurities
solvent extraction	moves product from water into an immiscible solvent
recrystallisation	purifies a solid (removes soluble and insoluble impurities)
drying agent	water from an organic liquid
distillation	separates by boiling temperature; collects a volatile product
steam distillation	separates a volatile, water-immiscible compound below 100 °C
melting / boiling temperature	checks identity and purity

MARK-SCHEME WORDING

- Solvent extraction: "the two solvents are immiscible / two layers are formed" and "the solute is more soluble in the extracting solvent" (WCH15 Oct 2025).
- Distillation: "the aldehyde is distilled off to prevent its further oxidation" (not "the temperature is too high to reach") (WCH15 Oct 2022).
- Anti-bumping granules: "so the solvent boils smoothly" (WCH15 Oct 2022 MCQ).
- Washing with NaOH or Na_2CO_3: the acid is "converted to an (ionic) salt that is more soluble in water" because the "ions are solvated by water" (WCH15 Jun 2022).
- Recrystallisation: "insoluble impurities are removed by filtering the hot solution" (WCH15 Jan 2026 MCQ).

Often asked as: MCQs on each technique; "suggest why" a step is done; name the technique for purifying a solid product; choose and justify a heat source.

COMMON MISTAKE

"Wash to remove impurities" without saying **which**: Na_2CO_3 removes **acidic** impurities.

Forgetting to **vent** the separating funnel when using carbonate.

Using lots of solvent in recrystallisation, or washing crystals with warm solvent: both dissolve product.

Stoppering the top of a reflux condenser.

Thinking an impurity **raises** the melting temperature. It lowers it and widens the range (Oct 2021 MCQ).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Purity from data**: a damp sample gives a mass too high and an apparent purity above the real value (WCH15 Oct 2024 MCQ; only a minority answered it correctly).
- **Chromatography**: R_f values from TLC or paper chromatography, and GC retention times, as purity checks (WCH15 Jan 2023, Jun 2025).

Quick checks

QUICK CHECK

Which sequence converts bromoethane into propanoic acid?

- aqueous NaOH, then acidified dichromate
- KCN in ethanol, then heat with dilute HCl
- ammonia, then $LiAlH_4$
- KOH in ethanol, then steam

Cyanide adds a carbon (CH_3CH_2CN), and acid hydrolysis turns the nitrile into CH_3CH_2COOH. NaOH then oxidation would give ethanoic acid, with only two carbons.

QUICK CHECK

Which product forms when CH_3MgI reacts with propanone and the mixture is then treated with water?

- propan-2-ol
- butan-2-ol
- 2-methylpropan-2-ol
- butanoic acid

A Grignard reagent with a ketone gives a tertiary alcohol: the CH_3 joins the carbonyl carbon, giving $(CH_3)_3COH$.

QUICK CHECK

Why is a primary alcohol oxidised under reflux to make a carboxylic acid, but distilled to make an aldehyde?

- reflux removes water
- reflux keeps the aldehyde in the flask to be oxidised further; distilling removes it as soon as it forms
- distillation is faster
- the aldehyde needs a higher temperature

The aldehyde has a lower boiling point than the alcohol and acid, so distillation carries it away before it meets more oxidising agent.

QUICK CHECK

In recrystallisation, why is the minimum volume of hot solvent used?

- to save solvent
- to dissolve the impurities
- so the solution is saturated and the product crystallises on cooling
- so the solution cools faster

With too much solvent, much of the product stays dissolved when cold and the yield falls, or no crystals form at all.

QUICK CHECK

A student's crystals melt between 118 °C and 124 °C; the data book value is 127 °C. What does this suggest?

- the crystals are pure
- the crystals are impure: impurities lower the melting point and widen the range
- the thermometer reads too high
- the compound has decomposed into a purer form

A pure substance melts sharply at its data book value. A low, broad melting range points to impurities, such as remaining solvent.

QUICK CHECK

Which statement about hazard and risk is correct?

- a hazard is the chance of harm
- a dilute solution of a corrosive substance has the same risk as the concentrated one
- risk depends on how much of a hazardous substance is used and how
- flammable liquids are a risk only when heated with electric heaters

A hazard is the potential to cause harm; risk is the chance that harm actually happens, which falls with smaller amounts, lower concentrations and good control measures.